Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Computer Networks

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS



Computer Networks

Understanding computer networks is essential in today's interconnected world:

- **Communication**: How devices exchange data across the internet
- **Protocol Stack**: The layered architecture that makes networking possible
- **TCP/IP**: The fundamental protocols powering the internet
- **Applications**: From web browsing to video streaming
- **Security**: Understanding vulnerabilities and protection mechanisms

We'll explore the **TCP/IP stack** and see how your computer communicates with the world!

What is a Computer Network?

A **computer network** is a collection of interconnected devices that can exchange data and share resources.

Key Components:

- **Nodes**: Devices connected to the network (computers, phones, servers, routers)
- **Links**: Physical or wireless connections between nodes
- **Protocols**: Rules governing how data is transmitted
- **Data**: Information being exchanged between nodes

Example: When you browse a website, your computer (node) connects through WiFi (link) using HTTP protocol to fetch data from a web server (another node).

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    - *Coverage*: Small geographic area (home, office, building)
    - *Examples*: Your home WiFi, office network
    - *Characteristics*: High speed, low cost, private ownership

Types of Networks

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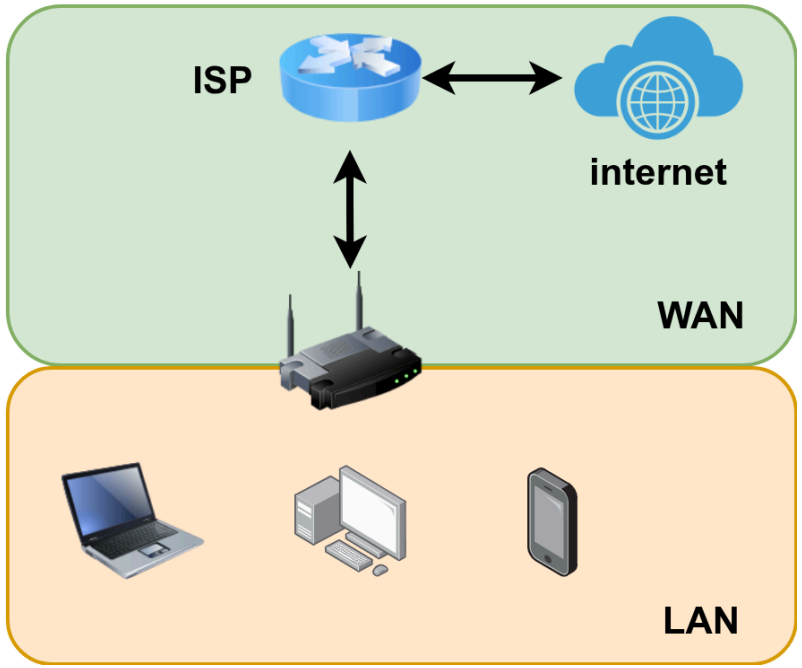
- Local Area Network (**LAN**)
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- Wide Area Network (**WAN**)
    - *Coverage*: Large geographic area (cities, countries, continents)
    - *Examples*: The Internet, corporate networks across multiple cities
    - *Characteristics*: Lower speed than LAN, uses public infrastructure

Types of Networks

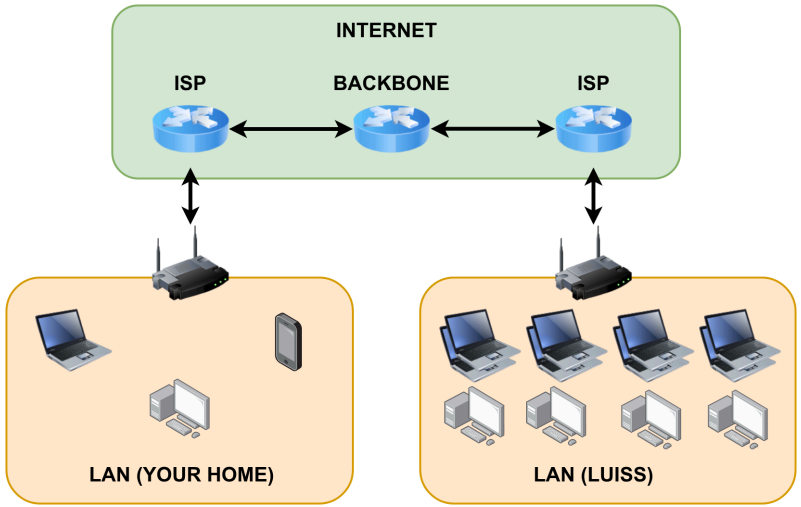
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    - *Examples*: The Internet, corporate networks across multiple cities
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- Other Types:
    - Metropolitan Area Network (**MAN**): City-wide coverage
    - Personal Area Network (**PAN**): Very small area (Bluetooth devices)


ISP
internet
WAN
LAN

The Internet: A Network of Networks

The **Internet** is the largest WAN, connecting billions of devices worldwide:



Internet Service Provider (ISP): Companies that provide internet access and maintain the infrastructure connecting networks together.

Why Do We Need Network Protocols?

Imagine trying to have a conversation where:

- Everyone speaks at once
- Messages arrive out of order
- Some words get lost in transmission
- No one knows who is speaking

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Protocols are standardized rules that solve these problems by defining:

- How to establish connections
- How to format and transmit data
- How to detect and correct errors
- How to manage network traffic
- How to terminate connections

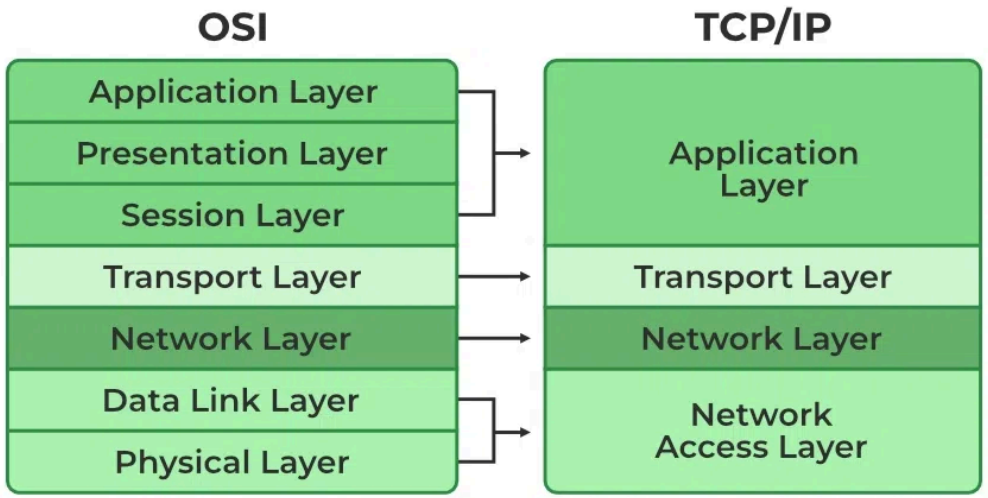
Without protocols, network communication would be **chaos**!

The OSI Model: A Conceptual Framework

The **OSI (Open Systems Interconnection)** model is a conceptual framework with **7 layers**:

Layer	Name	Function	Examples
7	Application	User-facing applications	HTTP, FTP, SMTP, DNS
6	Presentation	Data format translation, encryption	SSL/TLS, JPEG, MPEG
5	Session	Session management	NetBIOS, RPC
4	Transport	End-to-end communication	TCP, UDP
3	Network	Routing and addressing	IP, ICMP
2	Data Link	Physical addressing, error detection	Ethernet, WiFi, ARP
1	Physical	Raw bit transmission	Cables, fiber, radio waves

NOTE: The OSI model is **theoretical**. In practice, we use the **TCP/IP model**, simpler and more practical.

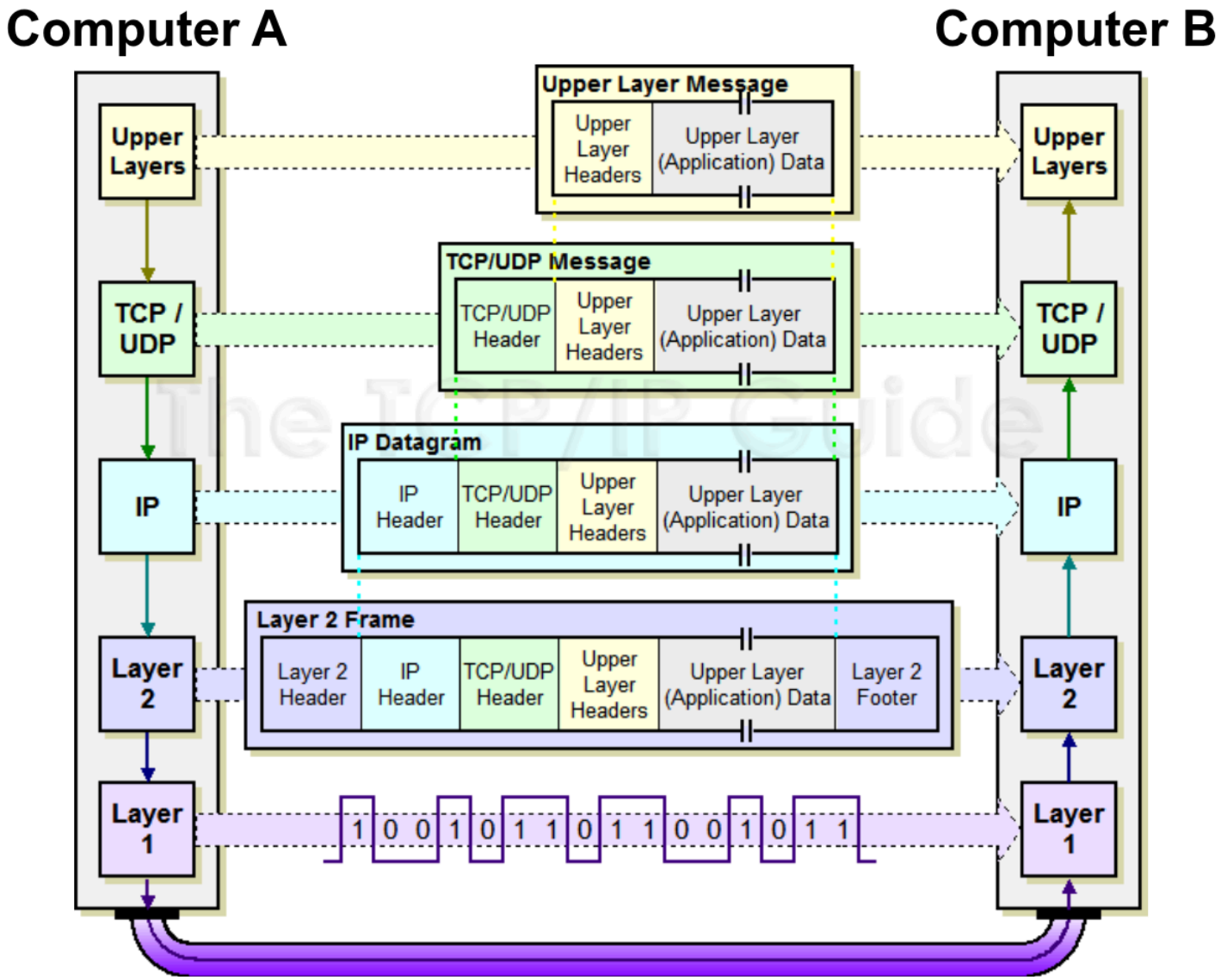

OSI
TCP/IP
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Application Layer
Transport Layer
Network Layer
Network Access Layer

The TCP/IP Model: The Internet's Foundation

The **TCP/IP model** has **4 layers** and is the actual protocol suite used on the internet:

Layer	OSI Equivalent	Function	Key Protocols
Application	5-7	User applications and services	HTTP, HTTPS, FTP, SMTP, DNS, SSH
Transport	4	End-to-end data delivery	TCP, UDP
Internet	3	Routing across networks	IP, ICMP
Network Access	1-2	Physical transmission	Ethernet, WiFi, PPP, ARP

Each layer adds its own **header** to the data, creating a **protocol stack**.


Computer A
Computer B
Upper Layers
TCP / UDP
IP
Layer 2
Layer 1
Upper Layer Message
Upper Layer Headers
Upper Layer (Application) Data
TCP/UDP Message
TCP/UDP Header
Upper Layer Headers
Upper Layer (Application) Data
IP Datagram
IP Header
TCP/UDP Header
Upper Layer Headers
Upper Layer (Application) Data
Layer 2 Frame
Layer 2 Header
IP Header
TCP/UDP Header
Upper Layer Headers
Upper Layer (Application) Data
Layer 2 Footer
1 0 0 1 0 1 1 0 1 1 0 0 1 0 1 1
Upper Layers
TCP / UDP
IP
Layer 2
Layer 1

Goals of each layer

- **Network Access Layer**: transfer data between devices on the same physical network
    - two devices connected to the same WiFi or Ethernet can communicate directly
- **Internet Layer**: route packets across multiple networks
    - transfer data from your home network to a web server across the internet
- **Transport Layer**: provide reliable or unreliable delivery of data between applications
    - allows multiple applications on the same device to communicate over the network
- **Application Layer**: enable user-level applications to communicate over the network
    - specific protocols for web browsing, email, file transfer, etc.

Layer 1: Network Access Layer

The **Network Access Layer** handles physical transmission of data:

- Converting digital data to electrical/optical/radio signals
- Physical addressing (MAC addresses)
- Accessing the physical medium (cables, wireless)
- Error detection at the hardware level
- Nodes are identified by their **MAC addresses**

MAC Address: A unique 48-bit identifier assigned to every network interface card (NIC)

- Example: `00:1A:2B:3C:4D:5E`
- Used for local network communication

Comparison of Physical Network Links

Technology	Type	Max Speed	Max Distance	Pros	Cons	Use Cases
Fiber Optic	Wired (Light)	100+ Gbps	100+ km	Very fast, immune to EMI, secure	Expensive, fragile	Backbone, data centers
Copper (Ethernet)	Wired (Electrical)	10 Gbps	100m	Cost-effective, easy setup	Limited distance, EMI-prone	LANs, home networks
WiFi (802.11)	Wireless (Radio)	1-10 Gbps	50-100m	No cables, mobile	Interference, security risks	Home, offices, hotspots
4G/5G	Wireless (Cellular)	100 Mbps - 10 Gbps	Several km	Wide coverage, mobile	Expensive, variable speed	Mobile, remote areas, IoT

EMI: Electromagnetic Interference

ADSL

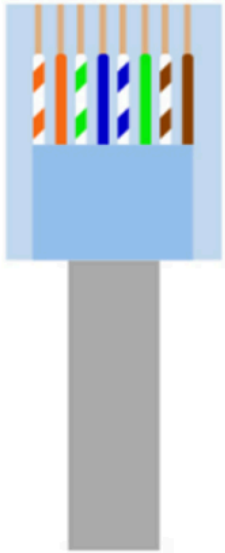
≈7mm





ETHERNET

≈11mm



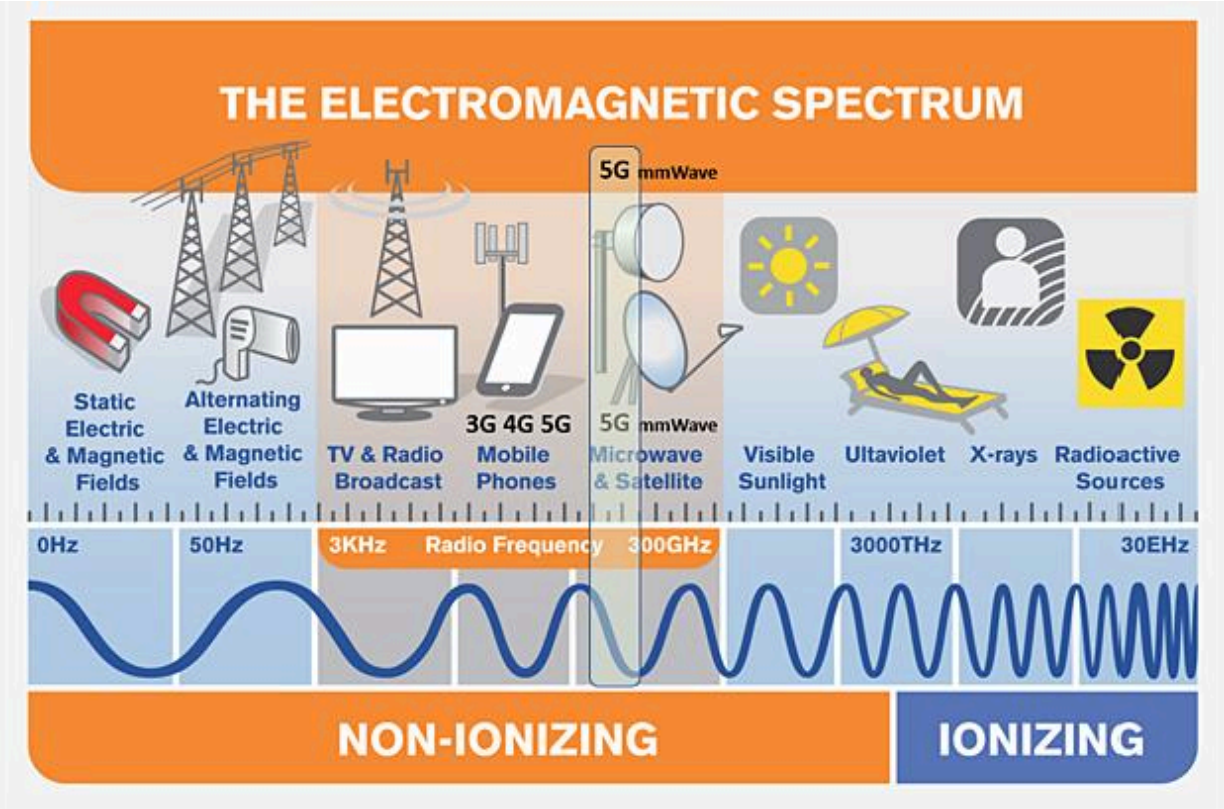


OPTIC FIBER

≈3mm






THE ELECTROMAGNETIC SPECTRUM
5G mmWave
Static Electric & Magnetic Fields
Alternating Electric & Magnetic Fields
TV & Radio Broadcast
3G 4G 5G Mobile Phones
5G mmWave Microwave & Satellite
Visible Sunlight
Ultaviolet
X-rays
Radioactive Sources
0Hz
50Hz
3KHz
Radio Frequency
300GHz
3000THz
30EHz
NON-IONIZING
IONIZING

Comparison of Wi-Fi Standards

	Wi-Fi 5	Wi-Fi 6	Wi-Fi 6E	Wi-Fi 7
Standard	IEEE 802.11ac	IEEE 802.11ax	IEEE 802.11ax	IEEE 802.be*
Transmit Frequency	5 GHz only	2.4 and 5 GHz	2.4, 5, and 6 GHz	2.4, 5, and 6 GHz
Max Number of Streams	8	8	8	16
Max Data Rate per Stream	866 Mb/s	1.2 Gb/s	1.2 Gb/s	2.9 Gb/s
Max Theoretical Data Rate	6.93 Gb/s	9.61 Gb/s	9.61 Gb/s	46.1 Gb/s
Typical Speed	1.3 Gb/s	5 Gb/s	5 Gb/s	18 Gb/s (preliminary)

*In development

Layer 2: Internet Layer

The **Internet Layer** is responsible for routing data across networks:

- Routes packets from source to destination
- Works across different types of networks
- Provides logical addressing (IP addresses)
- Key Protocol: **Internet Protocol (IP)**
- Nodes are identified by their **IP addresses**

We are using IPv4 but (slowly) moving to IPv6: extremely costly to change entire infrastructure!

IP Addressing

IP Address: A unique numerical identifier for each device on a network

- **IPv4**: 32-bit address (e.g., `192.168.1.1`)
    - Format: Four numbers (0-255) separated by dots
    - Problem: Only ~4.3 billion addresses (running out!)
- **IPv6**: 128-bit address (e.g., `2001:0db8:85a3:0000:0000:8a2e:0370:7334`)
    - Format: Eight groups of hexadecimal numbers
    - Solution: 340 undecillion addresses (virtually unlimited)

IP Addressing: Public vs Private

Public IP Addresses:

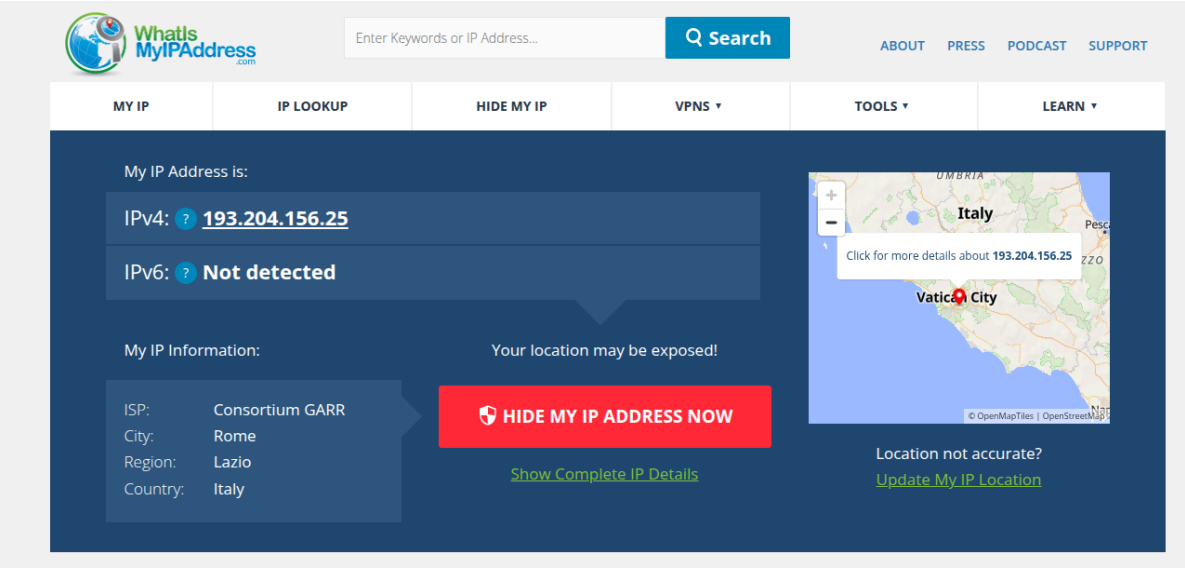
- Globally unique and routable on the internet
- Assigned by ISPs
- Required for direct internet communication
- Example: `8.8.8.8` (Google's DNS server)

Private IP Addresses:

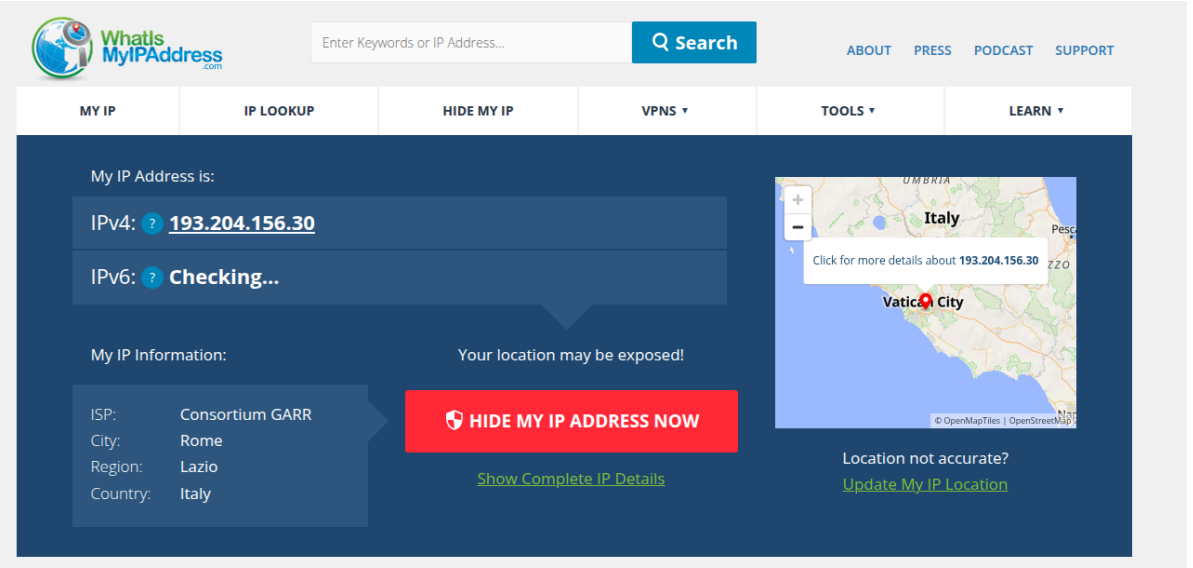
- Used within local networks (LANs)
- Not routable on the internet
- Can be reused in different networks
- Ranges:
    - `10.0.0.0` to `10.255.255.255`
    - `172.16.0.0` to `172.31.255.255`
    - `192.168.0.0` to `192.168.255.255`

To allow multiple devices with private IPs to share a single public IP address when accessing the internet, we use **NAT (Network Address Translation)** (see later).

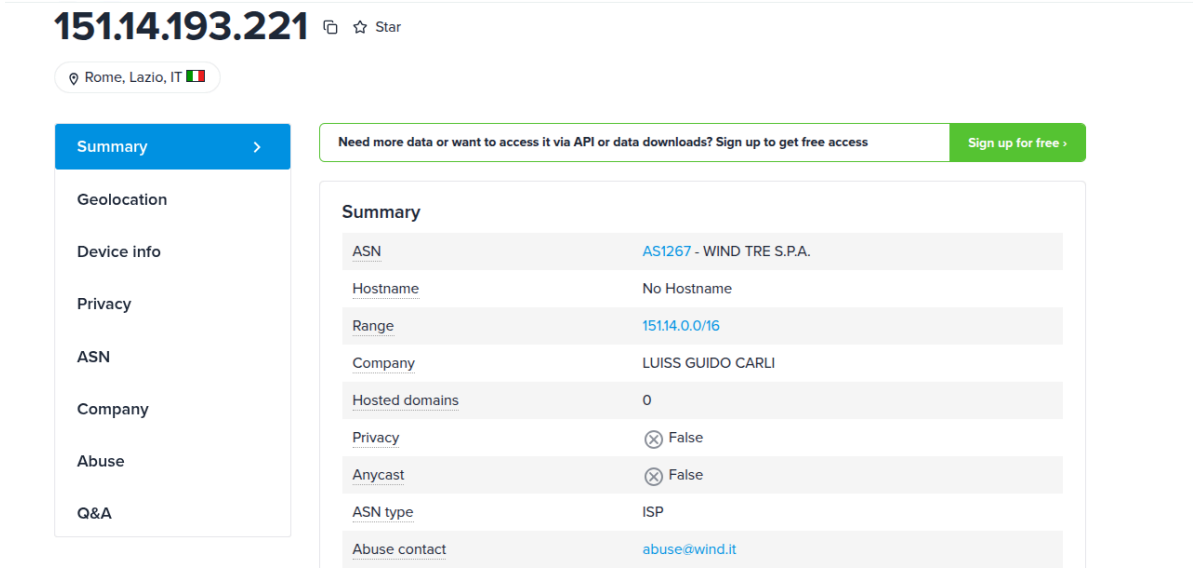
What is my IP in LUISS? [Ethernet]



What is my IP in LUISS? [LUISSWpa]



What is my IP in LUISS? [Eduroam]



Public IP ranges

Universities use the GARR network:

- Sapienza Università di Roma: from 151.100.0.0 to 151.100.255.255
- LUISS Guido Carli: 193.204.154.0 - 193.204.154.255 (also other ranges)

Some IP ranges used by italian ISPs:

- Vodafone Italia S.p.A. (AS30722): 5.88.0.0-5.95.255.255
- Fastweb SpA (AS12874): 93.32.0.0-93.47.255.255
- WIND TRE S.P.A. (AS1267): 151.48.0.0-151.63.255.255
- Telecom Italia S.p.A. (AS16232): 2.192.0.0-2.199.255.255

DHCP: Dynamic Host Configuration Protocol

DHCP automatically assigns IP addresses to devices on a network:

1. Device connects to network and broadcasts: "I need an IP address!"
2. DHCP server responds with available IP address and configuration
3. Device accepts and uses the assigned IP address for a lease period
4. When lease expires, device must renew or get a new address

What DHCP Provides:

- IP address: the unique identifier for the device
- Subnet mask: defines the network segment
- Default gateway (router): the path to external networks
- DNS servers: for domain name resolution (see later)
- Lease duration (typically 24 hours to 7 days)

Other Internet Layer Protocols

- **ICMP (Internet Control Message Protocol)**: Error reporting and diagnostics
  - Used by tools like `ping` and `traceroute`
  - Reports unreachable hosts, network congestion, timeouts
  - Example: "Destination Host Unreachable" error messages
- **ARP (Address Resolution Protocol)**: Maps IP addresses to MAC addresses
  - Operates between Internet Layer and Network Access Layer
  - When a device knows an IP address but needs the MAC address
  - Example: Router needs to know MAC address of `192.168.1.10` to deliver packet on local network

Layer 3: Transport Layer

Transport Layer provides end-to-end communication between applications through two main protocols:

- **TCP (Transmission Control Protocol)**:
  - Connection-oriented: Establishes connection before sending data
  - Reliable: Guarantees delivery, order, and error-free transmission
  - Flow control: Manages data rate to prevent overwhelming receiver
  - Use cases: Web browsing, email, file transfer (when reliability matters)
- **UDP (User Datagram Protocol)**:
  - Connectionless: No connection establishment
  - Unreliable: No delivery guarantees, packets may be lost or arrive out of order
  - Fast: Lower overhead, minimal delay
  - Use cases: Video streaming, online gaming, DNS queries.

TCP is more reliable but slower, while UDP is faster but less reliable. The choice depends on application requirements.

Moving from devices to applications

The Problem: Multiple applications can run on the same device simultaneously (web browser, email client, video call, etc.), each needing its own communication channel. How does the device know which incoming packet belongs to which application?

The Solution: **Ports** and **Sockets**

- **Port**: A numerical identifier (0-65535) assigned to a specific application or service
- **Socket**: The combination of an IP address and a port number, creating a unique endpoint for communication
    - Format: `IP:Port` (e.g., `192.168.1.10:54321`)
    - A socket uniquely identifies a specific application on a specific device

Ports: Addressing Applications

Port numbers identify specific applications or services on a device:

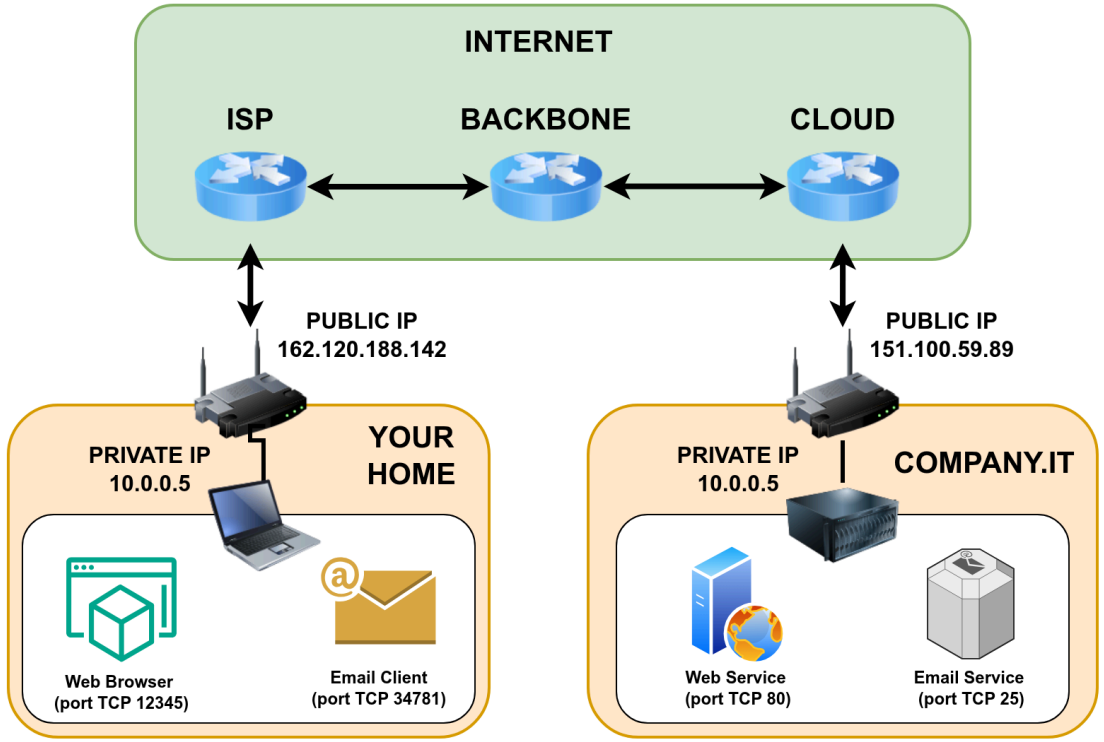
- Range: 0-65535
- Combined with IP: creates a unique *socket* (e.g., `192.168.1.1:80`)

Port Categories:

Range	Category	Usage
0-1023	Well-known ports	Standard services
1024-49151	Registered ports	Application-specific
49152-65535	Dynamic/Private ports	Temporary client connections

Common Well-Known Ports

- **80**: HTTP (web browsing)
- **443**: HTTPS (secure web browsing)
- **22**: SSH (secure remote access)
- **21**: FTP (file transfer)
- **25**: SMTP (email sending)
- **53**: DNS (domain name resolution)
- **3389**: RDP (Remote Desktop Protocol)



WHEN VISITING www.company.it: SRC=162.120.188.142:12345 DST=151.100.59.89:80
WHEN SENDING EMAIL FROM user@company.it: SRC=162.120.188.142:34781 DST=151.100.59.89:25

NAT (Network Address Translation) and Ports

NAT allows multiple devices with private IPs to share a single public IP address when accessing the internet. It exploits ports to differentiate between multiple internal devices.

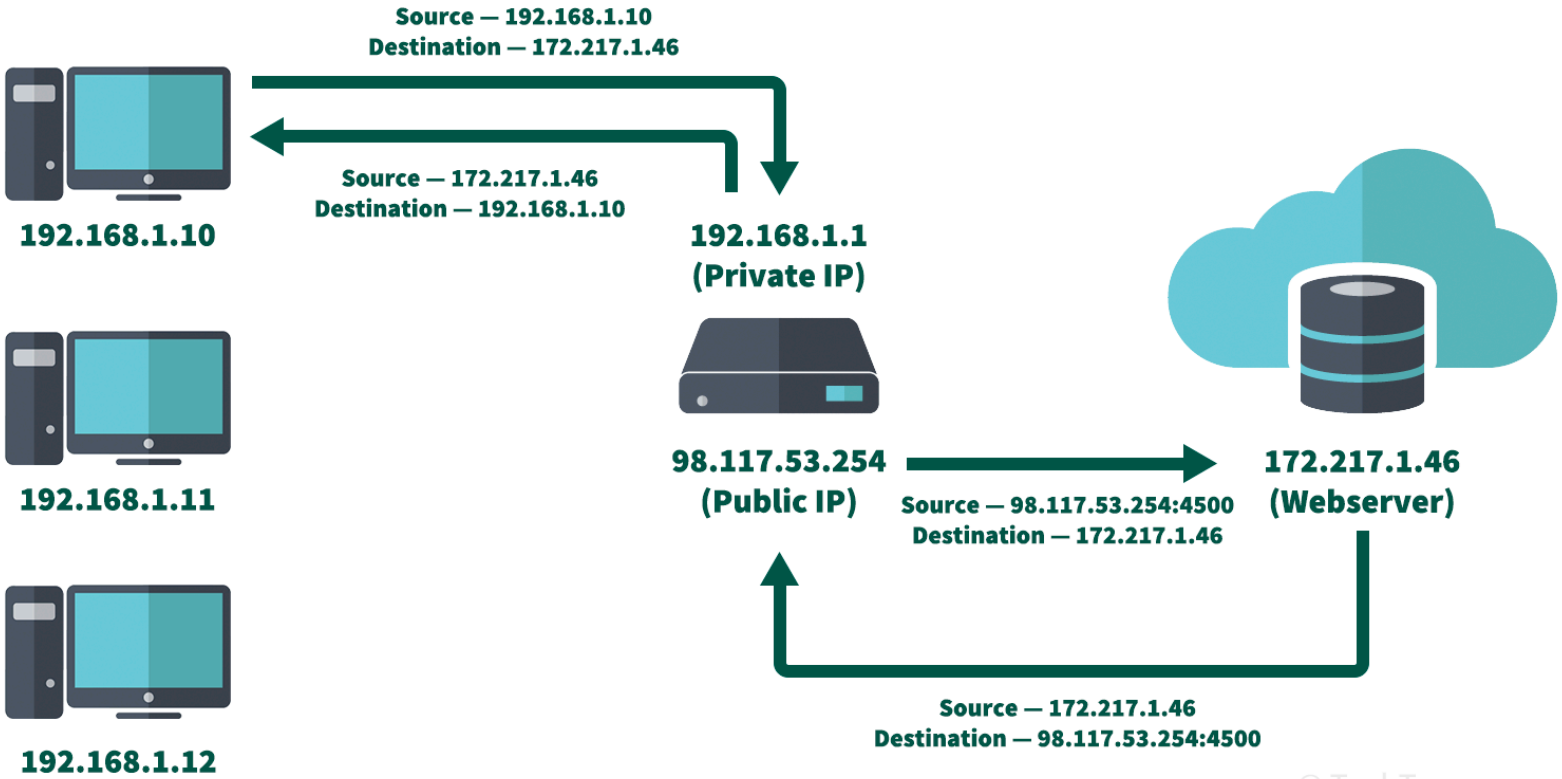
When a device sends a request to the internet:

1. The NAT device replaces the private IP address with its own public IP address.
2. It assigns a unique port number to the request.
3. The NAT device keeps track of the mapping between the internal IP address/port and the external IP address/port.

When a response comes back:

1. The NAT device uses the port number to identify the correct internal device.
2. It forwards the response to the appropriate internal IP address.

This way, multiple devices can share a single public IP address while maintaining separate connections.


Source — 192.168.1.10
Destination — 172.217.1.46
Source — 172.217.1.46
Destination — 192.168.1.10
192.168.1.10
192.168.1.11
192.168.1.12
192.168.1.1
(Private IP)
98.117.53.254
(Public IP)
Source — 98.117.53.254:4500
Destination — 172.217.1.46
172.217.1.46
(Webserver)
Source — 172.217.1.46
Destination — 98.117.53.254:4500
© TechTerms.com

Layer 4: Application Layer

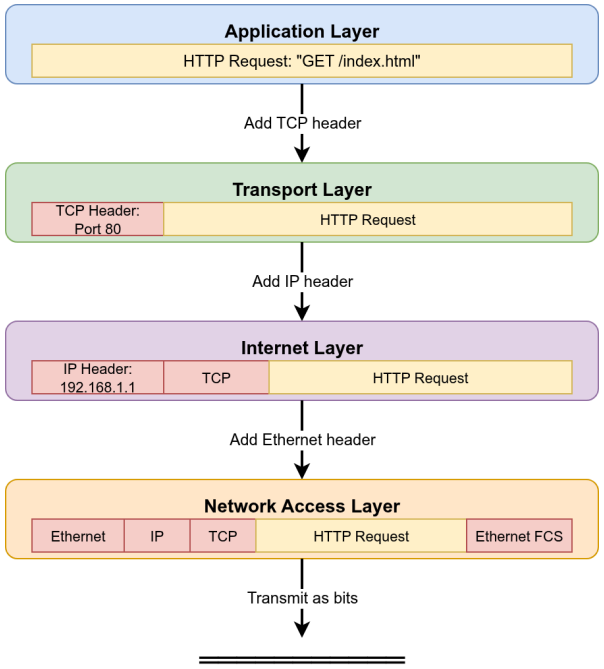
The **Application Layer** contains protocols that applications use directly:

- **HTTP/HTTPS** (Web Browsing)
    - HTTP: Hypertext Transfer Protocol (TCP port 80)
    - HTTPS: HTTP Secure with TLS/SSL encryption (TCP port 443)
    - Used for web pages, APIs, and web services
- **DNS** (Domain Name System)
    - Translates human-readable domain names to IP addresses
    - Reason why we do not need to remember IPs of websites!
    - Example: `www.google.com` → `142.250.185.46`
    - UDP Port 53

- **SMTP/POP3/IMAP** (Email when using an email client)
    - SMTP: Sending emails (TCP port 25, 587)
    - POP3: Retrieving emails, downloads and deletes (TCP port 110)
    - IMAP: Retrieving emails, keeps on server (TCP port 143)
- **FTP/SFTP** (File Transfer)
    - FTP: File Transfer Protocol (TCP ports 20, 21)
    - SFTP: Secure FTP over SSH (TCP port 22)

How Data Flows Through the Stack

When you send data, each layer adds its own **header** (encapsulation):



On the receiving side, each layer **removes** its header (decapsulation) and passes data up.

Example: Loading a Web Page

Let's trace what happens when you visit `https://www.example.com`:

1. **DNS Resolution** (finding the IP address):
    - **Your computer**:
        A. The user types `www.example.com` in the browser
        B. Browser checks if IP is cached locally; if not, sends DNS query
        C. **Application Layer**: DNS query "What is the IP address of www.example.com?"
        D. **Transport Layer**: Wraps query in UDP packet (UDP is fast, suitable for DNS)
            - Source Port: random high-numbered port (e.g., 54321)
            - Destination Port: 53 (DNS)
        E. **Internet Layer**: Adds IP header
            - Source IP: your computer's private IP (e.g., 192.168.1.10)
            - Destination IP: DNS server's IP (e.g., 8.8.8.8 - Google's DNS)
        F. **Network Access Layer**: Wraps in Ethernet/WiFi frame and sends to your router (default gateway)

- **Your router**:
    1. Receives the frame, extracts the UDP/IP packet
    2. **NAT**: Replaces your private IP with router's public IP and tracks the mapping
        - Source IP: router's public IP (e.g., 203.0.113.45)
        - Source Port: new port number for tracking (e.g., 12345)
    3. Forwards the packet to the internet

- **The Internet**:
    1. Multiple routers forward the packet hop-by-hop based on destination IP (8.8.8.8)
    2. Each router examines the IP header and forwards to the next router
- **DNS Server** (e.g., 8.8.8.8):
    1. Receives the UDP/IP packet on port 53
    2. Extracts the DNS query from the UDP payload
    3. Looks up `www.example.com` in its database
    4. Finds the IP address: `93.184.216.34`
    5. Builds DNS response packet with the IP address
    6. Sends UDP/IP packet back:
        - Source IP: 8.8.8.8, Source Port: 53
        - Destination IP: 203.0.113.45 (your router's public IP)
        - Destination Port: 12345 (the port NAT assigned)

- **The Internet**:
    1. Routers forward the response packet back to your router
- **Your router**:
    1. Receives the packet on port 12345
    2. **NAT**: Looks up the port mapping and translates back
        - Destination IP: 192.168.1.10 (your computer's private IP)
        - Destination Port: 54321 (original port)
    3. Forwards packet to your computer on the local network
- **Your computer**:
    1. Receives the UDP/IP packet on port 54321
    2. Extracts the DNS response: `www.example.com = 93.184.216.34`
    3. Browser now knows the server's IP address!

2. **HTTPS Connection and Request**:
    - **Your computer**:
        A. Browser initiates connection to `93.184.216.34` on port 443 (HTTPS)
        B. **Transport Layer - TCP 3-way handshake** (establishing reliable connection):
            - Your computer → Server: SYN (synchronize)
            - Server → Your computer: SYN-ACK (synchronize-acknowledge)
            - Your computer → Server: ACK (acknowledge)
        C. **Application Layer - TLS/SSL handshake** (establishing secure connection):
            - Negotiate encryption algorithms
            - Exchange certificates
            - Generate session keys
        D. **Application Layer**: Browser builds HTTPS GET request
            - `GET / HTTP/1.1`
            - `Host: www.example.com`

- **Your computer**:

    5. **Transport Layer**: Wraps in TCP segment
        - Source Port: random high-numbered port (e.g., 54322)
        - Destination Port: 443 (HTTPS)
    6. **Internet Layer**: Adds IP header
        - Source IP: 192.168.1.10 (your private IP)
        - Destination IP: 93.184.216.34
    7. **Network Access Layer**: Sends to router

- **Your router**:

    1. Receives the frame, extracts TCP/IP packet
    2. **NAT**: Translates addresses
        - Source IP: 203.0.113.45 (router's public IP)
        - Source Port: 23456 (new tracking port)
    3. Forwards to the internet

- **The Internet**:
    1. Routers forward the packet hop-by-hop to destination IP (93.184.216.34)
- **Web Server** (93.184.216.34):
    1. Receives TCP/IP packet on port 443
    2. Decrypts the HTTPS request using TLS session keys
    3. Processes the HTTP GET request
    4. Retrieves the web page content (HTML, CSS, JavaScript, images)
    5. Builds HTTPS response with the content
    6. Encrypts the response
    7. Sends TCP/IP packet back:
        - Source IP: 93.184.216.34, Source Port: 443
        - Destination IP: 203.0.113.45 (your router's public IP)
        - Destination Port: 23456 (NAT tracking port)

- **The Internet**:
    1. Routers forward the response back to your router
- **Your router**:
    1. Receives the packet on port 23456
    2. **NAT**: Translates back using port mapping
        - Destination IP: 192.168.1.10 (your computer)
        - Destination Port: 54322 (original port)
    3. Forwards to your computer
- **Your computer**:
    1. Receives TCP/IP packet on port 54322
    2. **Transport Layer**: TCP ensures data arrived correctly and in order
    3. **Application Layer**: Browser decrypts HTTPS response
    4. Browser parses HTML, loads CSS/JavaScript, renders images
    5. **You see the web page!**

NOTE: This is simplified. In reality, modern web pages require dozens or hundreds of additional requests for resources (images, scripts, stylesheets, fonts, etc.), each following a similar process.